

NAUCOM – COOKING RECIPE ANDROID APP

BUSINESS REQUIREMENT DOCUMENT

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TABLE OF CONTENTS

1. EXECUTIVE SUMMARY	3
2. PROJECT DESCRIPTION	3
3. PROJECT SCOPE	4
4. BUSINESS DRIVERS	5
5. PRESENT PROCESS	5
6. PROPOSED PROCESS	5
7. FUNCTIONAL REQUIREMENTS	6
7.1. PRIORITY RATING SYSTEM.....	6
7.2. REQUIREMENTS CATEGORIES (RC)	6
8. NON-FUNCTIONAL REQUIREMENTS	7
9. RESOURCES.....	7
10. SCHEDULE, TIMELINE, AND DEADLINES	8
11. ASSUMPTIONS & CONSTRAINTS.....	8
11.1. Constraints	8
11.2. Assumptions.....	8
12. GLOSSARY	9
13. REFERENCES	9
14. APPENDIX	9

1. EXECUTIVE SUMMARY

NauCom is an Android mobile application designed to help users discover cooking recipes based on available ingredients, plan meals, and manage their favourite dishes – with Vietnamese language support via Google Translate API integration.

This document defines the business requirements and functional specifications for the NauCom application, developed as a team academic project (3 members) under Agile/Scrum methodology.

Purpose of this BRD: This Business Requirements Document defines the business context, scope, stakeholders, functional and non-functional requirements, and success criteria for NauCom. It serves as the primary reference for development, testing, and project review.

Audience: Development team (3 members), academic supervisors, and any stakeholder reviewing project deliverables.

2. PROJECT DESCRIPTION

Many busy individuals and households struggle with daily meal decisions, often leading to food waste due to unused ingredients left in refrigerators. Furthermore, most high-quality international recipe databases (such as Spoonacular) are provided exclusively in English, creating a language barrier for non-English speaking Vietnamese users.

NauCom resolves these challenges by offering an intelligent ingredient-based recipe discovery engine, automatic translation into Vietnamese, personalized recipe bookmarking, and a community space for sharing cooking experiences and visual feedback.

3. PROJECT SCOPE

This section defines the boundaries of the NauCom project, outlining what is included and excluded from the current release scope.

IN-SCOPE ITEMS	OUT-OF-SCOPE ITEMS
User Authentication: Email/Password registration, login, profile management, and logout via Firebase Auth.	Integrated Grocery Delivery/ E-Commerce Cart.
Recipe Search Engine: Search by recipe name, dish category, or available pantry ingredients via Spoonacular API.	External Social Media Sharing integrations.
Recipe Detail Visualization: Detailed views for ingredients, step-bystep instructions, preparation time, and cloric metrics.	Push Notifications (e.g., automated meal reminders).
Favourites Management: Save, organize, and delete favourite recipes backed by Cloud Firestore	Offline Recipe Access & Caching.
Community Comments: Post text comments and image attachments under specific recipe listings.	Advanced Admin Moderation Portal.
Automated Translation: On-the-fly English-to-Vietnamese translation via Google Translate API.	Cross-platform support (iOS/ Web).

4. BUSINESS DRIVERS

Business Driver 1: Food Waste Reduction

Enable users to utilize available pantry ingredients efficiently through tailored recipe recommendations, reducing household food waste.

Business Driver 2: Meal Planning Efficiency

Streamline meal decision-making for busy individuals by consolidating recipe discovery into a single, intuitive platform.

Business Driver 3: Accessibility and Localization

Eliminate language barriers by translating international recipe databases into localized Vietnamese content.

Business Driver 4: Community Engagement

Foster user retention and interaction by enabling user-generated feedback, photos, and recipe commentary.

5. PRESENT PROCESS

- **Manual Search:** Users search fragmented websites, blogs, or social media platforms for ideas.
- **Language Barrier:** English-language recipes require manual translation of technical culinary terms and unit measurements.
- **Ingredient Inefficiency:** Users frequently purchase unnecessary items rather than utilizing ingredients already available at home.
- **Disorganized Storage:** Recipes are saved via scattered screenshots or temporary notes, making future retrieval difficult.

6. PROPOSED PROCESS

- **Input & Search:** Users open NauCom and input their available refrigerator ingredients.
- **API Query & Translation:** The application queries the Spoonacular API, retrieves matching recipes, and automatically translates content into Vietnamese.
- **Execution:** Users view nutritional metrics, prep time, and step-by-step cooking instructions.
- **Engagement & Storage:** Users save preferred recipes to their Favourites list and upload dish photos or reviews to the community section.

7. FUNCTIONAL REQUIREMENTS

7.1. PRIORITY RATING SYSTEM

VALUE	RATING	DESCRIPTION
1	Critical/ Immediate	Core dependency; essential for initial system launch.
2	High	High-value capability required for Minimum Viable Product (MVP).
3	Medium/ Moderate	Enhances user experience and operational completeness.
4	Low	Non-essential feature; nice- to-have functionality.
5	Future/ Prospective	Out of scope for current release; slated for future iterations.

7.2. REQUIREMENTS CATEGORIES (RC)

ID	REQUIREMENT	PRIORITY	RAISED BY
BR-01	The system shall enable secure user registration and login using Email and Password.	2 (High)	BA/ UC001, UC002
BR-02	The system shall allow users to update profile details and reset details and reset account passwords.	3 (Medium)	BA/UC003
BR-03	The system shall support recipe queries by dish name, meal category, or input ingredients.	2 (High)	BA/UC004
BR-04	The system shall render detailed recipe views including ingredients, instructions, cooking time, and calorie counts.	2 (High)	BA/UC005
BR-05	The system shall allow authenticated users to save, view, and remove recipes in a Favourites list.	3 (Medium)	BA/UC006
BR-06	The system shall support user commentary with text entries and image uploads under recipes.	3 (Medium)	BA/UC007
BR-07	The system shall automatically translate English recipe content into Vietnamese using the Google Translate API.	3 (Medium)	BA/Endpoint
BR-08	The system shall integrate with the Spoonacular API to retrieve live recipe data and nutritional information.	2 (High)	BA/ External System

BR-09	The system shall securely store user profiles, favourites, and comments on Firebase Firestore.	3 (Medium)	BA/Firestore
BR-10	The system shall gracefully handle network loss or API rate limits with user-friendly warnings.	2 (High)	BA/ System Stability

8. NON-FUNCTIONAL REQUIREMENTS

ID	CATEGORY	REQUIREMENT	ACCEPTANCE CRITERIA
NFR-01	Performance	API Response Time	Search queries and recipe details must render within 3.0 seconds under standard network conditions.
NFR-02	Security	Data Protection	Authentication credentials managed via Firebase Auth; plain-text passwords must never be stored.
NFR-03	Reliability	Fault Tolerance	The system must catch API timeouts or connectivity losses without crashing, displaying user feedback message
NFR-04	Usability	Localization	Recipe content must translate seamlessly into Vietnamese upon user request.
NFR-05	Compatibility	OS Support	The application must run stably on Android 8.0 (API Level 26) and higher.
NFR-06	Scalability	Rate Limit Management	The system must handle Spoonacular free-tier quota limits (150 points/day) without breaking core clients functions.

9. RESOURCES

- Human Resources:
 - 01 Project Lead/ Business Analyst: Requirements gathering, BRD/FS documentation, Sprint management.
 - 01 Android Frontend Engineer: Native UI/UX implementation and activity workflows in Android Studio.
 - 01 Backend & Integration Engineer: Firebase Auth/ Firestore implementation, Spoonacular & Translation API integration.
- Development Stack: Android Studio (Java), GitHub, Firebase Console, Figma.
- External Services: Spoonacular API, Google Translate API Endpoint, Firebase Cloud Services.

10. SCHEDULE, TIMELINE, AND DEADLINES

- Project Duration: 8 – 10 weeks (Agile/Scrum Framework).
- Key Milestones:
 - *Sprint 1 (Weeks 1 – 2):* BRD/FS Sign-off, Figma Wireframing, Project setup (Android Studio & Firebase)
 - *Sprint 2 (Weeks 3 – 4):* User Authentication (UC001, UC002) & Profile Management (UC003).
 - *Sprint 3 (Weeks 5 – 6):* Spoonacular API Integration, Recipe Search (UC004), Recipe Details (UC005), Translation Engine.
 - *Sprint 4 (Weeks 7 – 8):* Favourites Management (UC006) & Community Comments with Image Uploads (UC007).
 - *Sprint 5 (Weeks 9 – 10):* Integration Testing, NFR Optimization, Bug Fixing, Final Acceptance.

11. ASSUMPTIONS & CONSTRAINTS

11.1. Constraints

- Operations are bounded by the free tier limit of the Spoonacular API (150 requests/day).
- Fixed execution timeline of 8 – 10 weeks with a 3-member development team.
- No dedicated budget for paid translation microservices; dependent on free translation endpoints.
- Restricted exclusively to the Android operating system.

11.2. Assumptions

- End-users maintain active Internet connectivity while using the application.
- Spoonacular API provides reliable uptime and accurate nutritional data payload.
- Firebase Spark plan capacity is sufficient for initial project testing and evaluation.

12. GLOSSARY

TERM/ ABBREVIATION	EXPLANATION
BRD	Business Requirements Document – A formal document detailing business needs and project requirements.
API	Application programming Interface – Software intermediary allowing two applications to communicate.
Firebase Auth	Google cloud-based service for managing user authentication.
Firestore	Scalable NoSQL cloud database for storing realtime application data.
Spoonacular API	Third-party food and recipe API providing ingredient search and nutrition metrics.
MVP	Minimum Viable Product – Version of a product with just enough features to be usable by early customers.

13. REFERENCES

NAME	LOCATION/ LINK
Spoonacular API Documentation	https://spoonacular.com/food-api/docs
Firebase Android Setup Guide	https://firebase.google.com/docs/android/setup
Agile BRD Template Standard	Smartsheet Agile Business Requirements Documentation Framework
NauCom Use Case Diagram	https://github.com/rosemary134/business-analysis-portfolio/tree/main/03-Naucom-Cooking-Recipe-Android-App/Use-Case-Diagram
NauCom Wireframes (Figma)	bit.ly/naucomapp-figma
NauCom GitHub	https://github.com/rosemary134/naucomapp

14. APPENDIX

- **User Role Matrix:** All authenticated users share equal access rights to Search, View Details, Bookmark Favourites, and Post Comments.
- **API Quota Contingency:** If daily Spoonacular API limits are reached, the system will fall back to locally cached sample data or notify users to retry the following day.